

KOREAN ECONOMIC DEVELOPMENT

KDI School of Public Policy and Management

Changkeun Lee

Objectives



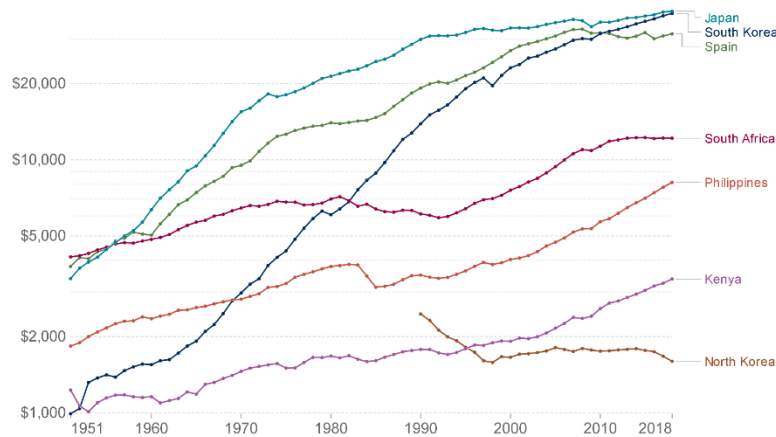
- **Overview of Korea's historical economic development**
- **Placing Korea in a comparative context**
- **Explaining emerging challenges after rapid economic growth**



Characters of Korean Economic Growth

GDP per capita, 1951 to 2018

This data is adjusted for inflation and for differences in the cost of living between countries.

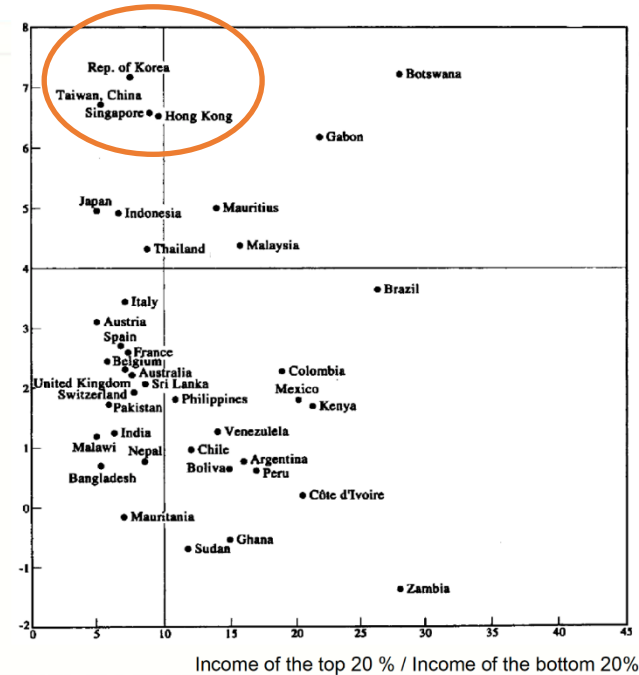


Source: Maddison Project Database 2020 (Bolt and van Zanden, 2020)
Note: This data is expressed in international-\$¹ at 2011 prices.

OurWorldInData.org/economic-growth • CC BY

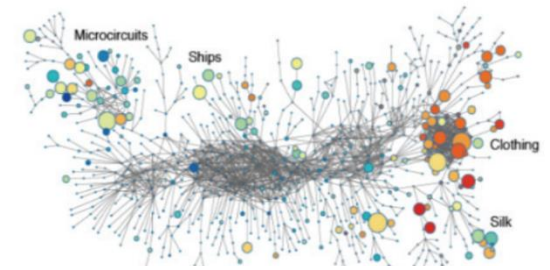
1. International dollars: International dollars are a hypothetical currency that is used to make meaningful comparisons of monetary indicators of living standards. Figures expressed in international dollars are adjusted for inflation within countries over time, and for differences in the cost of living between countries. The goal of such adjustments is to provide a unit whose purchasing power is held fixed over time and across countries, such that one international dollar can buy the same quantity and quality of goods and services no matter where or when it is spent. Read more in our article: 'What are Purchasing Power Parity adjustments and why do we need them?'

GDP Growth per capita (percent)



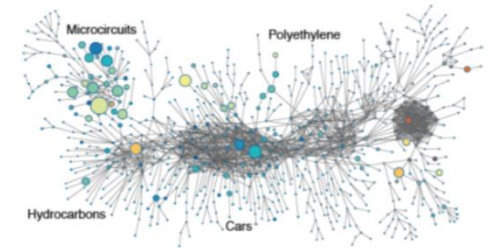
A

South Korea 1970s



B

South Korea 2000s



- Rapid growth
- Inclusive growth, as seen in other East Asian countries.
- Growth driven by product diversification



Comparative Context

		Structural transformation (<i>industrialization</i>)	
		Slow	Rapid
Investment in fundamentals (<i>human capital, institutions</i>)	Low	(1) No growth	(2) Episodic growth
	High	(3) Slow growth	(4) Rapid, sustained growth (South Korea, Taiwan, and Hong Kong)

- Dani Rodrik's typology - “one economics, many recipes”
- East Asian countries accomplished both economic growth, institutional improvement and structural transformation.
- Our question: what made it possible?



Drivers of Success

- **What people think:**

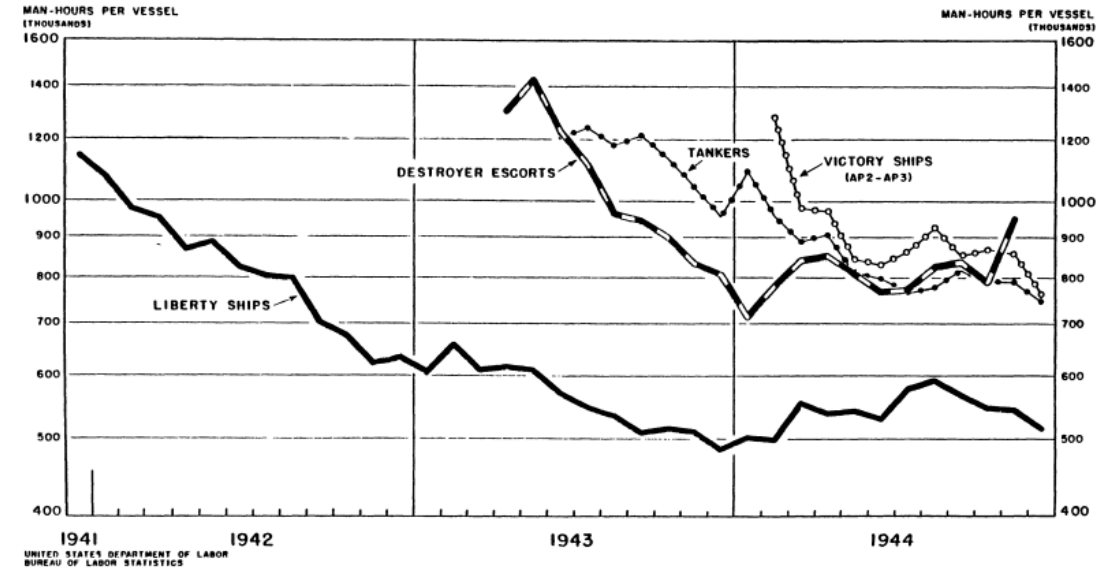


- **But academic research highlights public-private partnership**
 - Diagnosis of the changing global economic environment and industries
 - Shared vision and goals
 - Deliberation about policy goals and effective strategies
 - Coordination between government branches and public-private businesses
- **Economic research never advocates for any context-blind specific policy; rather, it emphasizes agility and adaptability.**



Sources of Economic Growth

Year	Aggregate GDP growth rate	Contribution to aggregate GDP					Labor income share
		Capital	Human capital	Employment	Labor hours	TFP	
1971–1979	8.47%	3.54%	0.42%	4.88%	−0.55%	0.17%	0.66
(1971–1982)	(7.44%)	(3.71%)	(0.29%)	(4.02%)	(0.00%)	(−0.58%)	(0.66)
1982–1989	9.48%	3.98%	0.44%	3.85%	−0.84%	2.05%	0.69
1989–1997	7.32%	3.22%	0.41%	2.76%	−0.47%	1.40%	0.74
(1989–1998)	(5.86%)	(3.02%)	(0.54%)	(1.82%)	(−0.58%)	(1.06%)	(0.74)
1998–2005	5.75%	1.61%	0.37%	1.91%	−0.20%	2.06%	0.74
(1999–2005)	(5.29%)	(1.60%)	(0.42%)	(1.91%)	(−0.68%)	(2.04%)	(0.74)
Total period	7.10%	3.15%	0.40%	2.97%	−0.37%	0.94%	0.70



- Capital investment was crucial in early growth
- Korea transitioned to TFP (innovation)-driven growth
- To economists, innovation means more **efficiency**. (process innovation)
- However, Rober Lucas explains with the example of US Liberty ship during WW2:
 - Learning effect increases efficiency. But it diminishes soon
 - **Introduction of new and better products** extends learning curve.
 - Insight: product and process innovations are not separate.
- Next question: how could Korea start the innovation engine?



Institution Building

- **Economists highlight the importance of inclusive institutions**
 - One should not take markets for granted.
 - Inclusive institutions: protection of **property rights** + **provision of public goods**.
 - Inclusive institutions encourage individuals to participate in productive activities
 - Both centralization and pluralism are required for individual institutions.
 - Extractive institutions tend to create a rent-seeking society.
- **Path dependence and history**
 - Good institutions are not always adopted: transaction costs + elite resistance
 - Without external shock or mutual agreement for transition, initial institutions tend to persist.
 - It took long for European countries to build inclusive institutions.
 - Many newly-independent countries after WW2 had their institutions transplanted
 - but had problems in solidifying them.



Institution Building

- **Legacy of Japanese colonialism**
 - (+) physical capital, elite human capital, modern institutions
 - (-) mass education, lack of experience
 - My view: foundations of modern Korean growth were formed after independence.
- **Constitutional period**
 - Power vacuum after independence from Japan
 - US military gov introduced US political & educational systems + conducted land reform with relatively less resistance
 - Cold war provided favorable conditions in trade & aid.
 - However, 1950's Korea was a rent-seeking society depending on US aid.
 - Since 1960s, Korea became a productive society.
 - Linked export to import & aid disbursement
 - Prove outside, then we will give support"

Major Chronological Events

1910 - Annexation of Korea by Japan
1941 - Start of the Pacific War
1945 - Japan's Surrender
Division along the 38th Parallel
US Military Government
1948 - KOR government established
1950-53 - Korean War
1960 - Syngman Rhee's Resignation
1961 - May 16 Military Coup

Institution Building and Outside Environment

- **The Cold War and Korea**

- Initially, the United States adopted a private market-led stance.
- Perceiving the threat from the Soviet Union, the U.S. shifted to state-led support for capitalist countries.
- The **Marshall Plan** aimed to restore the industrial capacities of Germany and Japan and pursue economic reconstruction in Europe and East Asia centered around these countries.
- The Korean government initially sought economic self-reliance but clashed with U.S. aid focused on essential goods.

- **Institutional Changes**

- Introduction of the separation of powers, the American-style party system, and egalitarian educational system; adoption of meritocracy principles in recruitment.
- Hostility towards Japanese imperialism and collaborators, felt the need for reform due to North Korea's institutional reforms.
- **Land reform** targeting land owned by Japanese – this was a precursor to large-scale reforms after the establishment of the government.
- Most newly independent countries faced difficulties in stabilizing early institutions.

Industrialization

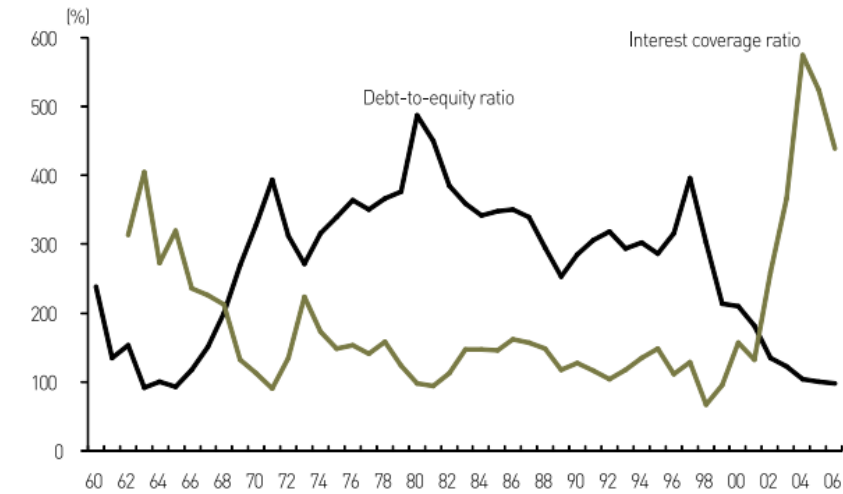
- **1950s: Export Promotion to Secure Foreign Currency**
 - Due to **foreign currency shortages**, Gov link imports to exports to secure foreign currency.
 - They kept Korean Won's value high to maximize the value of aid funds had a negative impact on exports.
 - Export promotion focused on primary products did not lead to industrial diversification.
- **1960s Export Promotion Policy**
 - Policies such as the **normalization of the exchange rate**, interest rate subsidies for export industries, and financial support promoted exports.
 - Government-led collection and sharing of information on overseas markets - Establishment of KOTRA.
 - Light industries based on low wages with comparative advantage were the main focus.
- **1970s Heavy and Chemical Industrialization**
 - After 1972, policies were pursued to strengthen national defense.
 - Gov promote **heavy and chemical industries**: POSCO (1970-73) and Hyundai Heavy Industries.
 - Strategic allocation of financial resources to the heavy and chemical industry sectors.
 - Establishment of coordinating bodies such as the Economic Planning Board and the Heavy and Chemical Industry Committee for policy coordination.

Industrial Policy

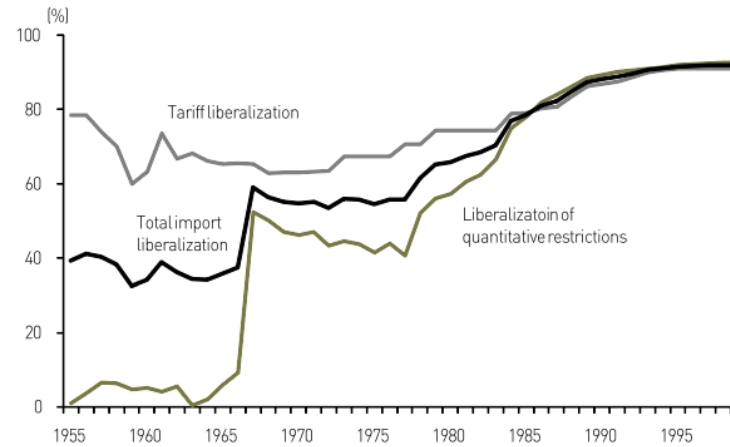
- **Windows of opportunity**
 - Created by changes in technology, market demand, regulations
 - Old leaders have much sunk costs; they have incentives to maintain old paradigm
 - Latecomer countries can skip some stages and government-firm's joint effort make it easier – **leapfrogging**.
- **Government's role**
 - Minimized direct intervention: focused on sharing goals, addressing externalities
 - Shared vision: **from imitation to innovation, trade-centered integrated approach**
 - Skill promotion and search/experimentation
 - Sharing information and exposing firms to external stimulus
 - Negotiated with foreign investors to achieve knowledge transfer
 - State guarantee on private sector foreign borrowing
- **Key characteristics: feedback mechanism and emphasis on agility**



1980s: Transition to a Stable and Market-Driven Economy



Note: Interest coverage ratio = operational profit / interest expense.
Source: Bank of Korea (<http://ecos.bok.or.kr>).

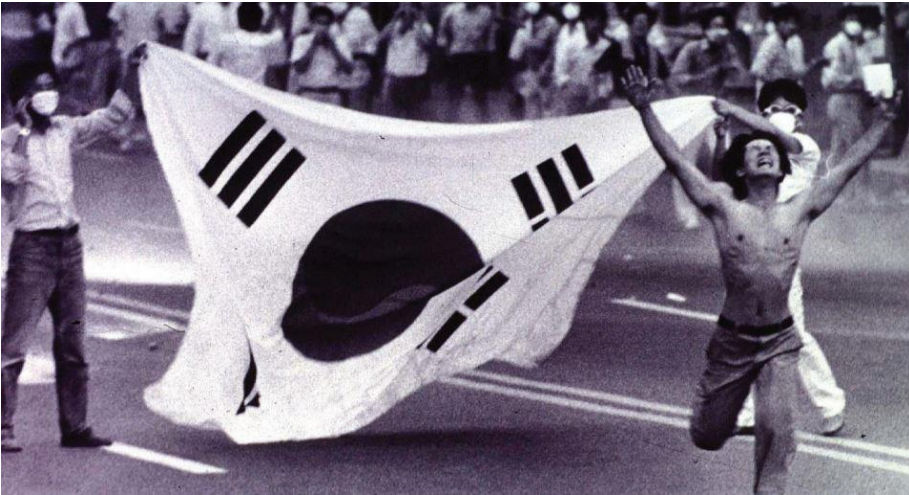


Note: 'Tariff liberalization' refers to the inverse of total tariff as defined in Table 2-4. 'Liberalization of quantitative restrictions' refers to the number of freely-imported items divided by the number of total items. 'Total import liberalization' is the average of these two ratios. Quantitative restrictions include those imposed not only by the Ministry of Commerce and Industry but also by other ministries.

Source: Kwang Suk Kim (1988), Table 5; Kwang Suk Kim (2001), Table IV-1.

- The economic recession caused by the oil shocks of the 1970s brought severe crises.
- Although a new military regime took power in the 1980s, there were drastic changes in economic policy.
- Shift to **market-driven policies**, implementation of **structural adjustments** for this transition.
- Tightening of monetary policy, restructuring of the financial sector and large enterprises, abandonment of protectionist policies.
- Policies to foster small and medium-sized enterprises (SMEs) for the diversification of the economy.

Democratization and Globalization



- **Democratization in 1987**

- The growth of the middle class increased demands for political democratization.
- Economically, it marked the **end of the low-wage system** – introduction of the minimum wage, etc.
- Political democratization led to liberalization and globalization.

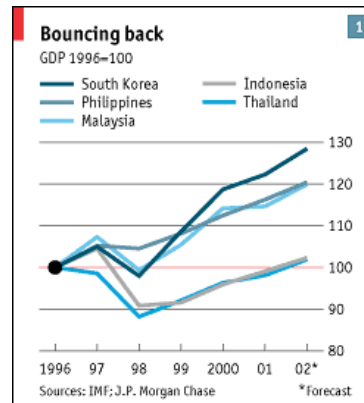
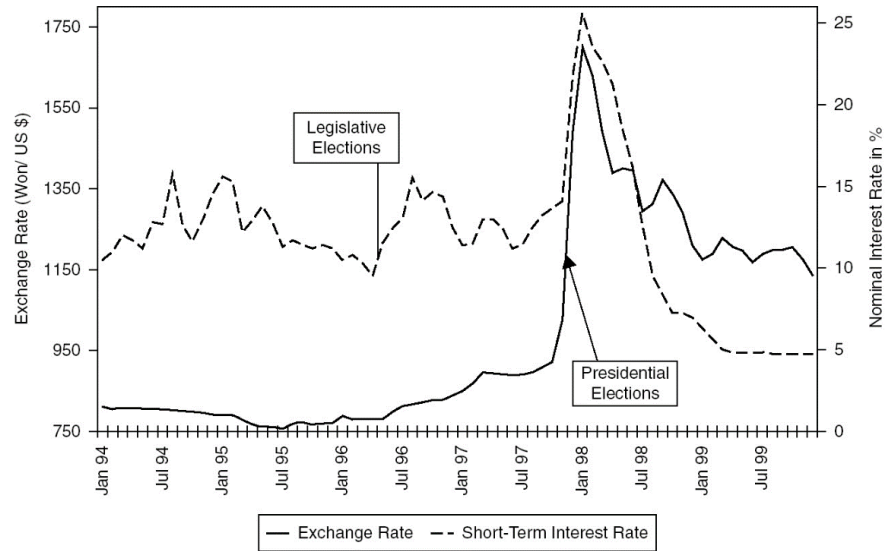


- **Globalization Efforts in 1990s**

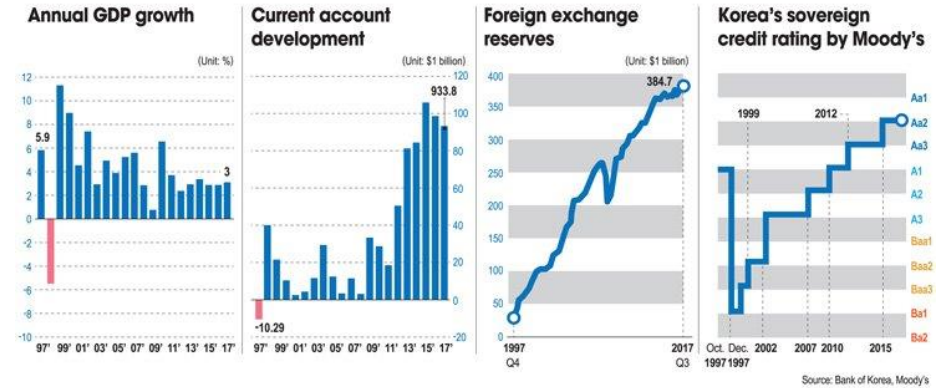
- Many companies invested in intangible assets or relocated production facilities overseas, striving to **improve productivity**.
- High-productivity companies continuously adapted to changes in the global market – leading to **increased inequality**.
- Financial market liberalization led many industrial capitals to enter the financial sector and borrow from abroad.



Asian Financial Crisis of 1997 and Reforms



What has happened to Korea since 1997 financial crisis



Causes

- In practice, a fixed exchange rate system was operated – vulnerable to speculative attacks.
- With the opening of the financial market, long-term borrowing and short-term lending practices became established, while borrowing-based investment behavior continued.

Responses: Four Major Reforms

- Corporate restructuring, labor market reform, public sector reform, financial market reform.
- The government intervened to ensure companies focused on core business areas, reduced the scale of the public sector, and cleaned up insolvencies in the financial market.
- The end of the lifelong employment practice in the labor market – strong external shocks forced reforms that were difficult in the past.

Emerging Challenges in a New Century

- **China Syndrome?**

- The rise of China provided both advantages and threats to Western advanced countries.
 - The U.S. lost between 1999 and 2011, 2.0–2.4 million jobs (Acemoglu et al., 2016).
- South Korea and Japan are prime examples of countries that benefited from China's rise.
- This is because they actively integrated into and built the **Global value chain**.
- During this period, Samsung and Hyundai Motor emerged as true global companies.

- **Demands for Work-Life Balance and Changing Policy Trends**

- In 2006, the workweek reduction to 40 hours –people no longer worked on Saturdays.
- Despite much opposition at the time, labor productivity increased – companies' proactive responses and gradual adaptation.
- As inequality increased, government policies gradually shifted from being industry-oriented to focusing on welfare.
- The structure where small and medium-sized enterprises (SMEs) follow the model set by the government and large corporations.
- This structure also became established in other areas where global large corporations set standards, and the government supports adaptation.



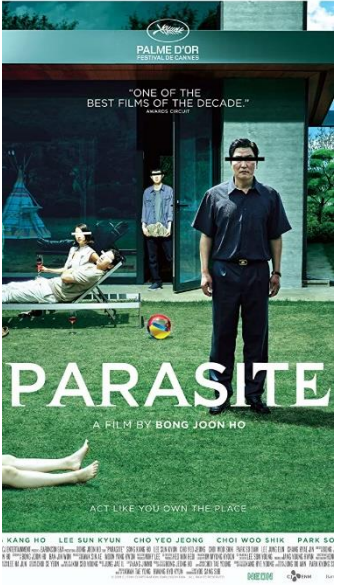
Korean Economic Development and the Business



- Economics emphasizes the importance of the market, but for repetitive tasks and strategic tasks, corporate **organization** is more efficient.
- It is especially important that knowledge is preserved and passed down within the organization.
- Samsung Electronics initially sold rice, but later diversified into food, wool, and electronics.
- Although criticized as an "octopus" strategy, this demonstrates the importance of a company's adaptability and management capabilities not confined to a sector.
- This is not a phenomenon unique to Korea – it is also commonly observed in the United States, Germany, and Japan where managers in one field achieve success in other industries.



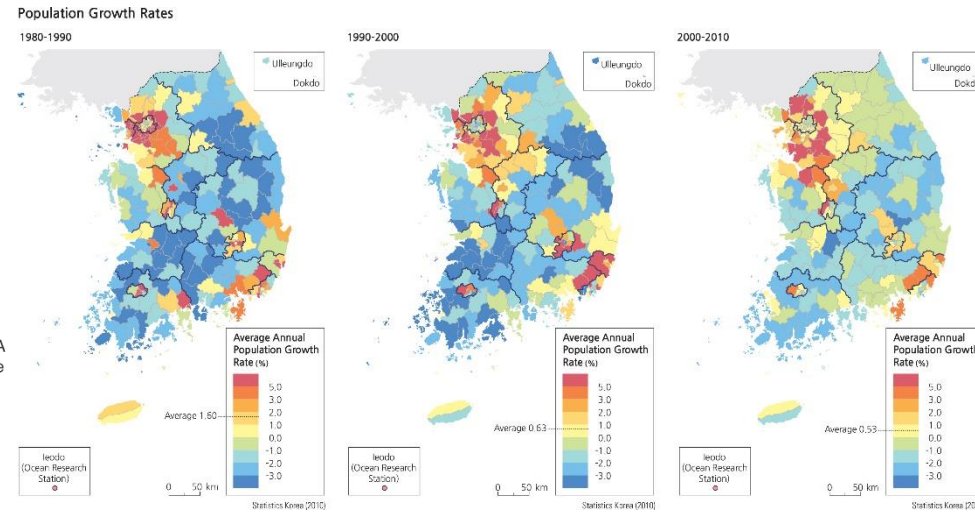
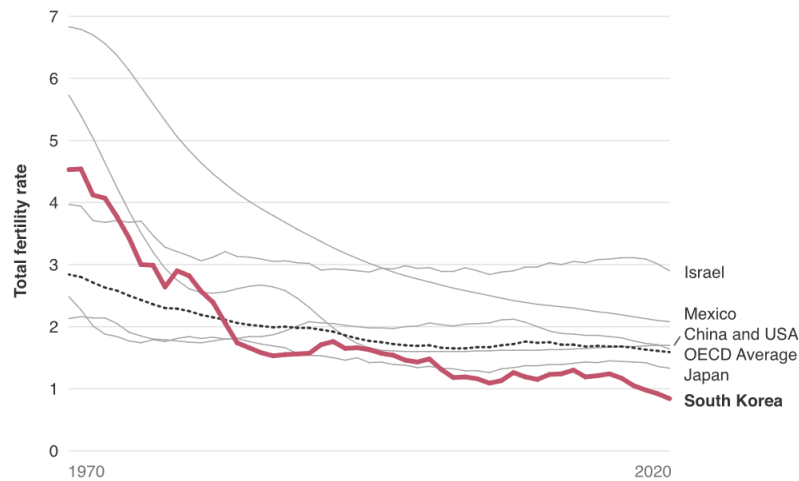
K-Culture: Korean Exceptionalism?



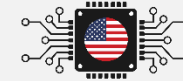
- **South Korea achieved success in the content sector rather than in traditional performing arts.**
- **This is because the content sector has high marketability and low cultural barriers.**
- **K-Culture and public policy**
 - Demand side: Economic growth and reduced working hours have increased demand for culture.
 - Supply side: Large companies' entry and government support for the pure arts sector
 - Opening the market to Japanese cultural products also ultimately contributed to the maturity of demand.
 - Intellectual property protection due to globalization increases the profitability of cultural products.
- **The growth of South Korea's cultural industry can be explained by universal principles rather than as an exceptional case.**



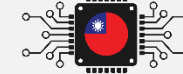
Objects in Mirror Are Closer Than They Appear



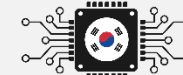
The Chip 4 Alliance: Players and their Strengths



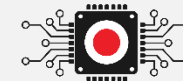
The United States - The country holds all Electronic Design Automation (EDA) tools licenses, and controls the market through tech giants such as Apple and other big tech companies.



Taiwan - more than half of the world's semiconductor chips are made here.



South Korea - home to Samsung, a giant in semiconductor manufacturing.



Japan - raw materials and equipment for semiconductor production come from here.

- **Demographic change**
 - Lowest fertility rate among OECD countries - economic and cultural factors
 - Population aging puts pressure on pension and labor markets
 - Population concentrated in Seoul metropolitan area – end of regions?
- **Global supply chain realignment**
 - Semiconductor, batteries, renewable energy
 - More security considerations mean that Korea needs to rebuild its global supply chain.
 - IPEF shows a new trade order based reorganizing supply chains.

